FINM3013 Introduction to Financial Derivatives

Chapter 10

Class exercises

Multiple Choice Questions

1. When dividends increase with all else remaining the same, which of the following is true?
   A. Both calls and puts increase in value
   B. Both calls and puts decrease in value
   C. Calls increase in value while puts decrease in value
   D. Puts increase in value while calls decrease in value

2. The price of a stock, which pays no dividends, is $30 and the strike price of a one year European call option on the stock is $25. The risk-free rate is 4% (continuously compounded). Which of the following is a lower bound for the option such that there are arbitrage opportunities if the price is below the lower bound and no arbitrage opportunities if it is above the lower bound?
   A. $5.00
   B. $5.98
   C. $4.98
   D. $3.98

3. Which of the following is NOT true? (Assume that present values are calculated from the end of the life of the option to today.)
   A. An American put option is always worth less than the present value of the strike price
   B. A European put option is always worth less than the present value of the strike price
   C. A European call option is always worth less than the stock price
   D. An American call option is always worth less than the stock price

4. Which of the following is true when dividends are expected?
   A. Put-call parity does not hold
   B. The basic put-call parity formula can be adjusted by subtracting the present value of expected dividends from the stock price
   C. The basic put-call parity formula can be adjusted by adding the present value of expected dividends to the stock price

D. The basic put-call parity formula can be adjusted by subtracting the dividend yield from the interest rate

5. Interest rates are zero. A European call with a strike price of $50 and a maturity of one year is worth $6. A European put with a strike price of $50 and a maturity of one year is worth $7. The current stock price is $49. Which of the following is true?
   A. The call price is high relative to the put price
   B. The put price is high relative to the call price
   C. Both the call and put must be mispriced
   D. None of the above